Section 4 Independence of events

- Independence of events
- Bernoulli's formula and binomial probability formula

Independence of events

Examples 1.5.1: There are 5 balls in the box (3 green and 2 red), take one ball randomly twice time with return it back. Let A={take a green ball at the first time}, B={take a green ball at the second time}. Please calculate P(B) and P(B|A).

Answer:

Obviously, P(B)= 3/5 and P(B|A)=3/5.

This means the occurrence of A has no impact on the occurrence of B.

$$P(B|A) = P(B) \Leftrightarrow P(AB) = P(A)P(B)$$

Independence of events

Definition 1.5.1: Any two events A and B are independent events if and only if $P(A|B) = P(A)$. Otherwise the events A and B are dependent.

$$\boldsymbol{P(B|A) = P(B) \Leftrightarrow P(AB) = P(A)P(B)}$$

Note: Event A and event B are independent, this means that the occurrence of event A is irrelevant to the probability of occurrence of event B.

Independence of events

The relationship between independence and disjoint.

Independence P(AB)=P(A)P(B)

$\not\Downarrow$ $\not\Uparrow$

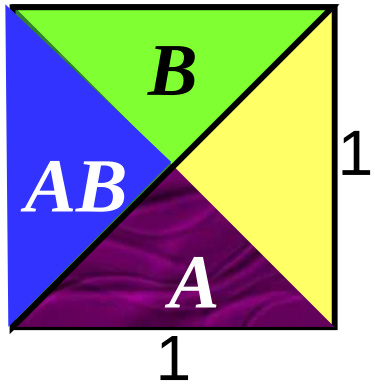
Disjoint $AB = \Phi$

$$P(A \cup B) = P(A) + P(B)$$

They are completely different concept.

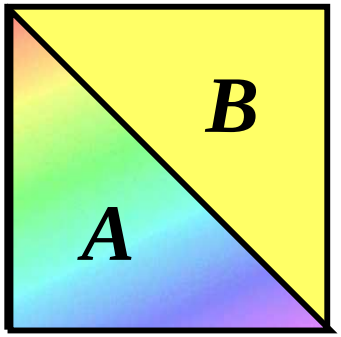
NO necessary connection!

Examples



$$P(A)=\frac{1}{2}, P(B)=\frac{1}{2},$$

$$P(AB)=P(A)P(B).$$



$$P(A)=\frac{1}{2}, P(B)=\frac{1}{2}$$

$$P(AB)=0,$$

$$P(A)P(B)=\frac{1}{4},$$

$$P(AB)\neq P(A)P(B)$$

Properties of independence

(1) Inevitable event Ω and impossible event Φ are independent of any event A.

$\Omega A = A$, $P(\Omega)=1$

$P(\Omega A) = P(A) = 1 \cdot P(A) = P(\Omega)\, P(A)$

→ Independent

$\Phi A = \Phi$, $P(\Phi)=0$

$P(\Phi A) = P(\Phi) = 0 = P(\Phi)\, P(A)$

→ Independent

Properties of independence

(2) If event A and B are independent, then we have

① A and $\overline{B}$ are independent

② B and $\overline{A}$ are independent

③ $\overline{A}$ and $\overline{B}$ are independent

Proof:

① $\because A = A\Omega = A(B+\overline{B}) = AB + A\overline{B}$

$$\therefore \quad P(A) = P(AB) + P(A\overline{B})$$

$$P(A\overline{B}) = P(A) - P(AB)$$

$$= P(A) - P(A)P(B)$$

$$= P(A)[1 - P(B)] = P(A)P(\overline{B})$$

same for B and $\overline{A}$

Independence of events

③ $\overline{B}$ and $\overline{A}$ are independent.

$$\because \overline{A}\,\overline{B} = \overline{A \cup B}$$

$$\begin{aligned}
\therefore \quad P(\overline{A}\,\overline{B}) &= P(\overline{A \cup B}) \\
&= 1 - P(A \cup B) \\
&= 1 - [P(A) + P(B) - P(AB)] \\
&= 1 - [P(A) + P(B) - P(A)P(B)] \\
&= [1 - P(A)] - P(B)[1 - P(A)] \\
&= [1 - P(A)] \cdot [1 - P(B)] \\
&= P(\overline{A})P(\overline{B}).
\end{aligned}$$

Independence of events

Example 1.5.3: X and Y are shooting the same enemy plane at the same time. Suppose that the probability of X shooting the enemy plane is 0.6, and the probability of B is 0.5. What is the probability of the enemy plane being damaged?

Answer:

A={X shoot the plane}, B={Y shoot the plane}, C={the plane being damaged}

$$\because \quad \overline{C} = \overline{A \cup B} = \overline{A}\,\overline{B}$$

$$\begin{aligned} \therefore \quad P(C) &= 1 - P(\overline{C}) = 1 - P(\overline{A})P(\overline{B}) \\ &= 1 - [1 - P(A)][1 - P(B)] \\ &= 1 - (1 - 0.6)(1 - 0.5) = 0.8 \end{aligned}$$

Independence of events

Note:

A set of events $\{A_1, A_2, \cdots, An\}$ is mutually independent

the set of events $\{\overline{A_1}, \overline{A_2}, \cdots, \overline{A_n}\}$ is mutually independent .

Independence of events

Definition 1.5.2: Definition of three independent events.

Three events A, B and C are independent

ONLY *if*

$$P(AB) = P(A)P(B)$$
$$P(AC) = P(A)P(C)$$
$$P(BC) = P(B)P(C)$$

And

$$P(ABC) = P(A)P(B)P(C).$$

Independence of events

Definition 1.5.3: Definition of *n* mutually independent events.

The set of events $\{A_1, A_2, \cdots, A_n\}$ is called mutually independent if for every subset $\{A_{i1}, A_{i2}, \cdots, A_{ik}\}$ $(k \geq 2)$ of $\{A_1, A_2, \cdots, A_n\}$ satisfies

$$\boldsymbol{P(A_{i1} A_{i2} \ldots Aik) = P(A_{i1})P(A_{i2}) \cdots P(A_{ik})}$$

If any two events in the set of events $\{A_1, A_2, \cdots, A_n\}$ is independent, i.e. for any $1 \leq i < j \leq n$, $P(A_i A_j) = P(A_i)P(Aj)$, then we call that any two events in set of events $\{A_1, A_2, \cdots, A_n\}$ are mutually independent to each other.

Independence of events

Example 1.5.3: There are four card with the same shape, there are a number serial on each card, which is 110, 101, 011, 000 respectively. Taking one card randomly, Let A_i ={the ith number on the taken card is 1} (i=1,2,3), proving:
(1)Any two of A_1, A_2 and A_3 are independent of each other.
(2)A_1, A_2 and A_3 are NOT mutually independent.

Answer: (1) $P(A_1)=\frac{2}{4}=\frac{1}{2}=P(A_2)=P(A_3)$

$$\because \quad P(A_1A_2)=\frac{1}{4}=P(A_1)P(A_2) \quad P(A_1A_3)=\frac{1}{4}=P(A_1)P(A_3)$$

$$P(A_2A_3)=\frac{1}{4}=P(A_2)P(A_3)$$

$$(2)\because \quad P(A_1A_2A_3)=\frac{0}{4}=0\neq P(A_1)P(A_2)P(A_3)=\frac{1}{8}$$

Independence of events

One important application: when a set of events $\{A_1, A_2, \cdots, An\}$ is mutually independent, calculating <u>at least one of $\{A_1, A_2, \cdots, An\}$ occurs</u> or <u>at least one of $\{A_1, A_2, \cdots, An\}$ does not occur</u>.

$$\boldsymbol{P(A_1 \cup A_2 \cup \cdots \cup A_n)} = 1 - \boldsymbol{P(\overline{A_1 \cup A_2 \cup \cdots \cup A_n})}$$

$$= 1 - \boldsymbol{P(\overline{A}_1 \overline{A}_2 \cdots \overline{A}_n)}$$

$$= \boldsymbol{1 - P(\overline{A}_1) P(\overline{A}_2) \cdots P(\overline{A}_n)}$$

For example: A machine has 100 parts, the status of each parts is mutually independent, any one of the parts does not work will cause the machine to stop working, what is the probability of the machine can work normally?

■ Bernoulli's formula and binomial probability formula

Definition 1.5.4: Let $\{E_i\}$ $(i = 1, 2, \ldots)$ is a series of random trials, the sample space of E_i is Ω_i, let A_k be any event in E_k, $A_k \subset \Omega_k$, if the probability of A_k occurrence does not depend on the results of other tests E_i (i $\neq k$), then $\{E_i\}$ is a mutually independent random trial sequence, referred as independent trial sequence.

■ Bernoulli's formula and binomial probability formula

Definition 1.5.5: If an experiment E only has two outcomes A and $\bar{A}$, we call E a Bernoulli trial. If $P(A) = p$, then $P(\bar{A}) = 1 - p$. If we repeat experiment n times independently, we call this n Bernoulli trials. n Bernoulli trials is a kind of independent trial sequence.

For example: Tossing a fair coin n times; rolling a dice to see if number 1 is on the top, repeat it n times.

■ Bernoulli's formula and binomial probability formula

Theorem: If the probability of event A occurs in Bernoulli trial is p ($0< p <1$), then the probability of event A happens k times in n Bernoulli trials is:

$$p_n(k) = C_n^k p^k (1-p)^{n-k}$$

And, $\sum_{k=0}^{n} p_n(k) = 1$.

$$\underbrace{A\,A\cdots A}_{k}\ \ \underbrace{\bar{A}\,\bar{A}\cdots\bar{A}}_{n-k},$$

Let X = k be the happening times of event A, $p_n(k) = C_n^k p^k (1-p)^{n-k}$ is called the **Binomial Distribution**, referred as $X \sim B(k; n, p)$

■ Bernoulli's formula and binomial probability formula

Example 1.5.4: There are 10 multiple choice questions in a test, each question has 4 choices, one of which is the correct answer. there is a candidate knows 6 questions of the test, she picks randomly choice for the rest questions, what is the probability that she picks *m* (*m*=0,1,2,3,4) right answers for the questions that she don't know?

Answer:

Let B_m={picking *m* right answer}, then

$$P(\mathrm{B}_m) = C_4^m (\frac{1}{4})^m (\frac{3}{4})^{4-m}$$

$$P(\mathrm{B}_0) = 0.316 \qquad P(\mathrm{B}_1) = 0.4219$$
$$P(\mathrm{B}_2) = 0.211 \qquad P(\mathrm{B}_3) = 0.048$$
$$P(\mathrm{B}_4) = 0.0039$$

HOMEWORK

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